



Section 8

Redshift Serverless

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Agenda

- Serverless
- Architecture
- Capacity in Serverless
- Feature parity in Redshift Serverless
- Check Capacity Consumption
- Migrate from Provisioned to Serverless



What is Serverless

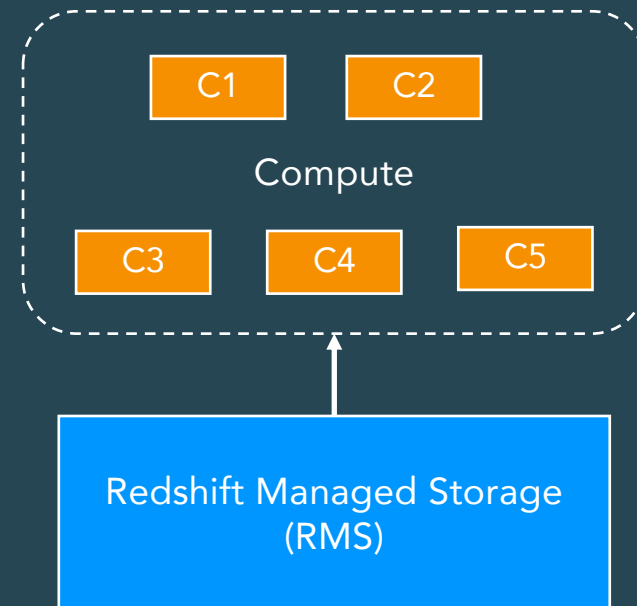
- Does it mean Redshift does not use any servers ???
 - NO !!
- We, as users, do not have to worry about the underlying infrastructure. Redshift will automatically take care of it.
 - Infrastructure – Compute (CPU + RAM)
- Scalability – Min & max capacity.
- High availability – Across multiple AZs.
- When to use serverless.
 - Zero overhead of infrastructure.
 - Unpredictable workload.
 - Uneven utilization.

Serverless Capacity

- RPU – Redshift Processing Unit.
- 1 RPU – 16GB memory and approx. 2 vCPUs.
- Base Capacity – Min RPUs for query processing.
 - Min base capacity – 8
 - Max base capacity – 512
- Scaling in increments of 8. 8 → 16 → 24 → 32 ...

Serverless Architecture

- Serverless compute.
- Workgroup – Collection of compute resources.
 - Memory percentage.
 - Concurrency scaling.
 - No. of RPU.
- Namespaces – Collection of Database objects, Users, Groups, Roles etc.
 - Storage related.
 - Encryption.
 - Datashares.



Features in RS Serverless

- Distribution Key & Sort Key. VACUUM, ANALYZE.
- Compression.
- COPY and UNLOAD commands.
- DMS, Zero-ETL, Data API.
- Materialized view.
- Streaming ingestion with MV.
- Query federation.
- Data sharing.
- WLM (**Automatic** and ~~Manual~~) – Query queues, concurrency scaling, SQA, QMR.
- Security

Hands-on (Capacity Consumption)

- Monitoring
- COPY command
- UNLOAD command
- Zero-ETL integration
- Table maintenance
 - VACUUM
 - ANALYZE
- Query optimization exercise
- Spectrum usage
- Real-time ingestion using Auto Refresh Materialized Views

Migrate from Provisioned to Serverless

- Take snapshot from provisioned cluster.
- Restore snapshot to serverless namespace.
- Audit logging – /aws/redshift/<namespace>/<log_type>
- Hands-on



Thank You

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